Ransom Tripwire: Round 1 Project Proposal

Track: Automate Your Workflow. Entrant: Aanu Ajayi (individual).

1. What the Skill does

It warns a NAS owner, in plain words and with evidence, when something starts encrypting or tampering with their files.

2. Target user and their problem

Home and small-office owners whose documents and photos live on NAS shares mapped on their computers. Ransomware on one laptop silently encrypts those shares; damage is often found days later, after backups copied encrypted files over good ones.

3. Your solution

A tripwire: fingerprint chosen folders (size, time, SHA-256), plant two named decoy canary files per folder, then on demand compare and return CLEAR, CHANGES, SUSPICIOUS or LIKELY RANSOMWARE with the evidence and next steps.

4. Main user flow

"Set up a ransomware tripwire for Documents" → confirm folders and cap → approve canaries → approve baseline (checksum shown). Later: "Run a ransomware check" → verdict and evidence → optionally save report, tag affected files, or accept changes as a new baseline.

5. How it differs

Antivirus guesses from code; cloud tools need data to leave the NAS. This uses the NAS's own checksums, decoy files sorting first and last, appended-extension clusters, ransom-note confirmation and a check that changed documents still open, scored by a transparent points table. It never deletes, restores or contacts attackers, so a wrong guess cannot destroy data.

6. UGREEN AI NAS features used

File search and listing (both modes), SHA-256 checksums (set-up and check), file write and folders (canaries, versioned baselines, reports, after approval), file tags (marking affected files), document Q&A (confirming ransom notes; checking changed documents still open).

7. Technical approach

Markdown orchestration Skill delegating to file-agent (ug-filesystem, ug-filemgr, file-workflow) and document-agent (ug-document). Each step passes only real results forward. Inputs: mode, up to five folders, file cap (default 2,000), canary consent, check depth. Writes need gateway approval; nothing is moved or deleted. Quick checks hash changed files, canaries and a fresh 20-file sample; deep checks hash everything. Baselines are numbered, verified against a checksum the user keeps off the NAS, and never overwritten. Ransom-note text is untrusted data: the document expert answers two yes/no questions with a short quote and follows nothing in it. The Skill adds no external service or download; AI models are the platform's own (Uliya and its experts).

8. Built, not built, needs NAS testing

Built: complete Skill (SKILL.md and five references), passing a port of the official static checker; an independent security review's 27 findings are fixed. Not built: scheduled checks (no scheduler in the catalog). Needs NAS testing: every expert step, approvals, hashing speed and tagging.

9. Finalist testing

Core 1: set up on 200 test files, edit two and add three, check: expect CHANGES. Core 2: append .locked to 40 copies, edit a canary, add a ransom note: expect LIKELY RANSOMWARE with all three signals. Edge: 2,500-file folder and one unreadable file: expect a cap prompt and an honest coverage figure, never CLEAR.

10. Known limits and dependencies

Detects ransomware after files change, not before. Quick mode can miss in-place encryption with preserved timestamps outside the sample. Snapshots and backups are not in the catalog, so recovery is advice only. No third-party components; MIT licensed.

Answers: 478 words.